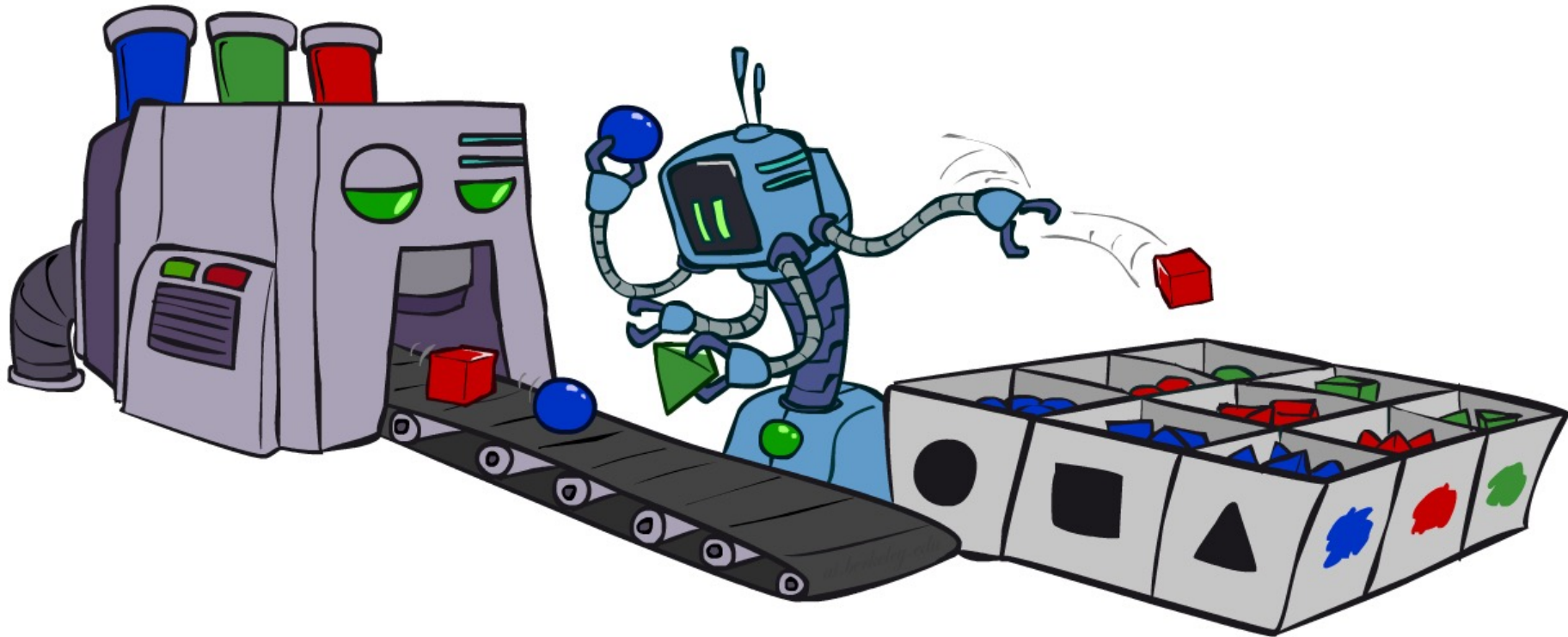


CS 188: Artificial Intelligence

Bayes' Nets: Sampling



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[These slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All CS188 materials are available at <http://ai.berkeley.edu>.]

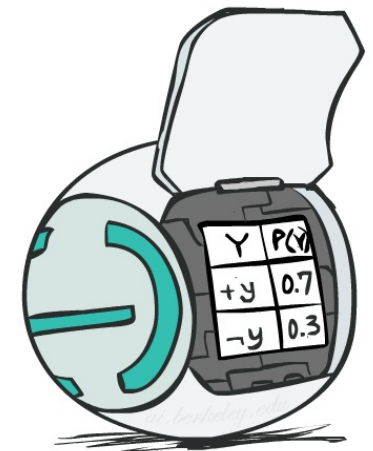
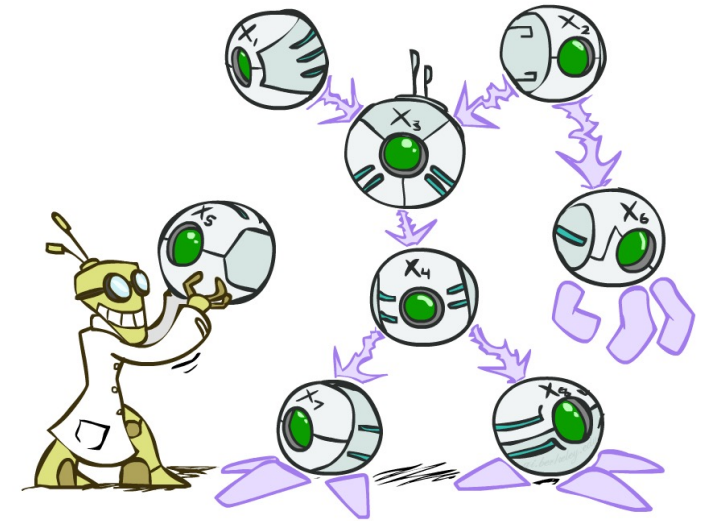
Bayes' Net Representation

- A directed, acyclic graph, one node per random variable
- A conditional probability table (CPT) for each node
 - A collection of distributions over X , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

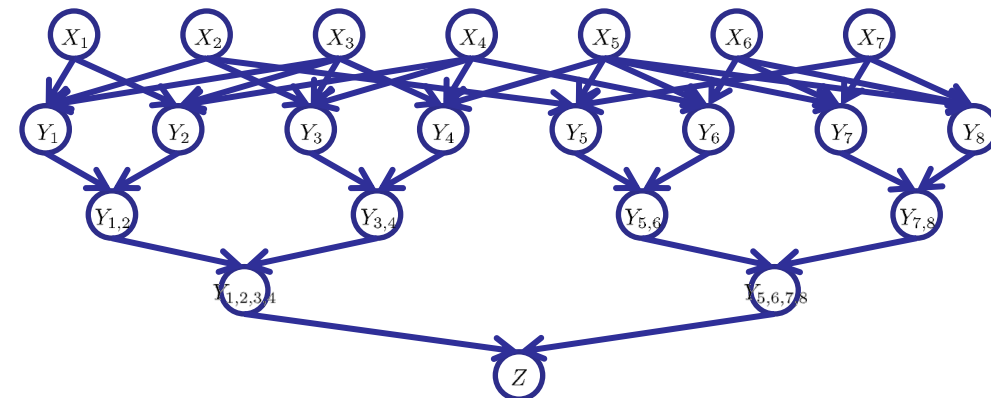
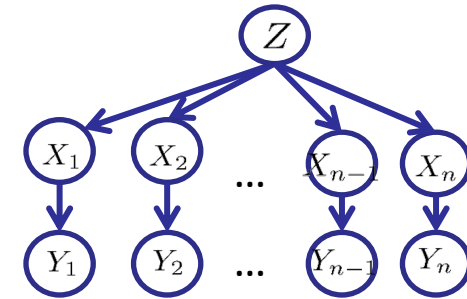
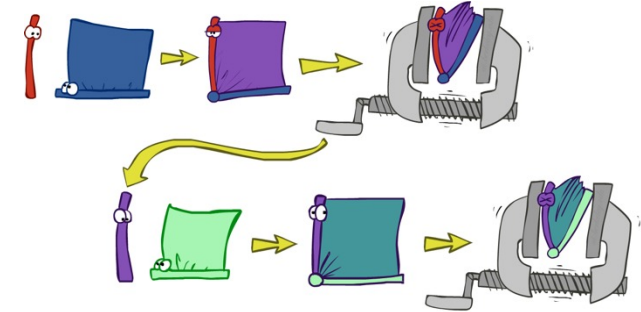
- Bayes' nets implicitly encode joint distributions
 - As a product of local conditional distributions
 - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

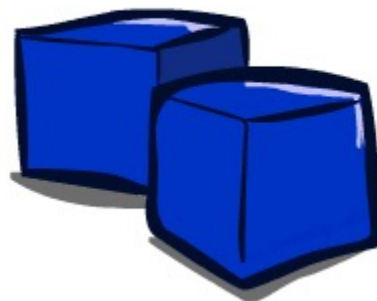


Variable Elimination

- Interleave joining and marginalizing
- d^k entries computed for a factor over k variables with domain sizes d
- Ordering of elimination of hidden variables can affect size of factors generated
- Worst case: running time exponential in the size of the Bayes' net



Approximate Inference: Sampling



Sampling

- Sampling is a lot like repeated simulation

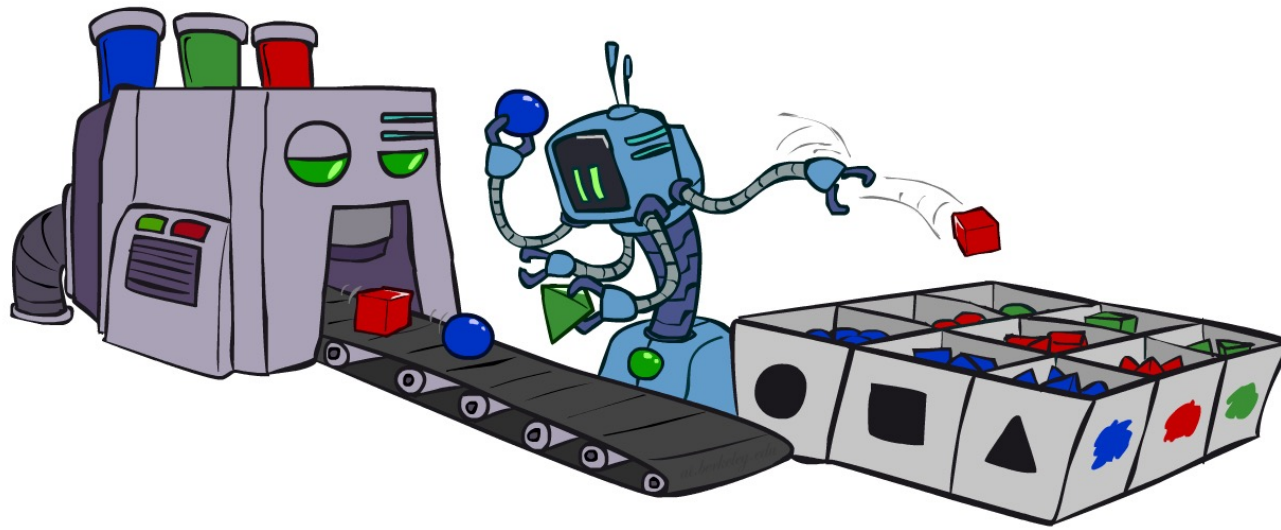
- Predicting the weather, basketball games, ...

- Basic idea

- Draw N samples from a sampling distribution S
 - Compute an approximate posterior probability
 - Show this converges to the true probability P

- Why sample?

- Learning: get samples from a distribution you don't know
 - Inference: getting a sample is faster than computing the right answer (e.g. with variable elimination)



Sampling

- Sampling from given distribution

- Step 1: Get sample u from uniform distribution over $[0, 1)$
 - E.g. `random()` in python
- Step 2: Convert this sample u into an outcome for the given distribution by having each outcome associated with a sub-interval of $[0,1)$ with sub-interval size equal to probability of the outcome

- Example

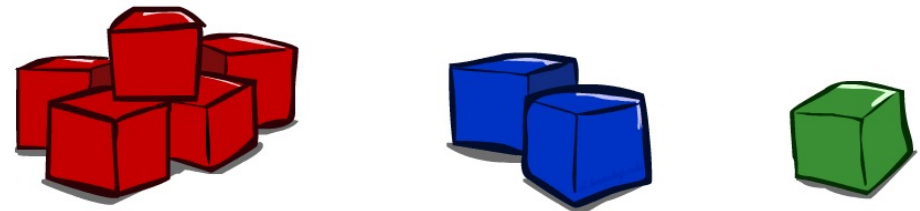
C	P(C)
red	0.6
green	0.1
blue	0.3

$$0 \leq u < 0.6, \rightarrow C = \text{red}$$

$$0.6 \leq u < 0.7, \rightarrow C = \text{green}$$

$$0.7 \leq u < 1, \rightarrow C = \text{blue}$$

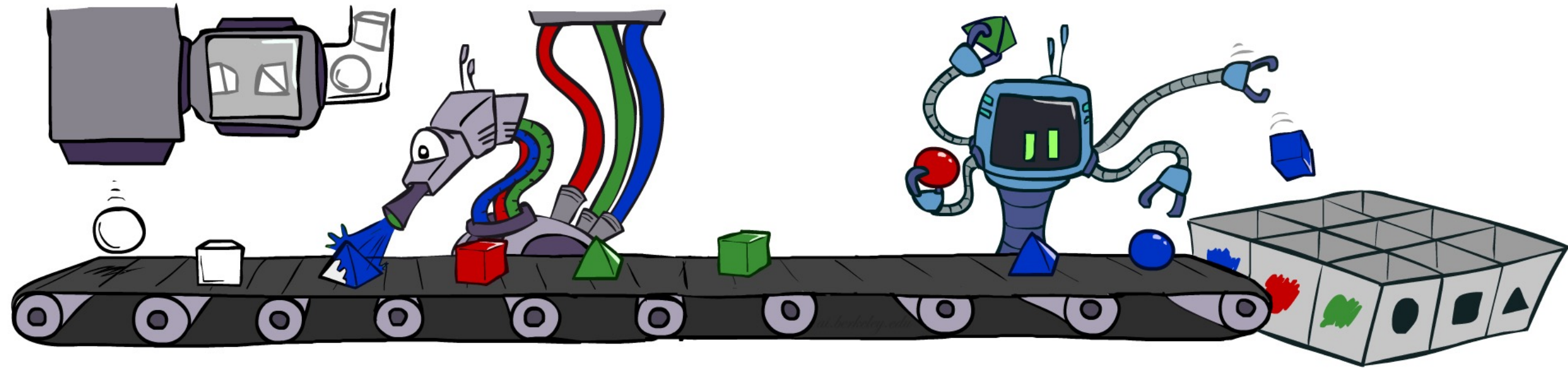
- If `random()` returns $u = 0.83$, then our sample is $C = \text{blue}$
- E.g, after sampling 8 times:



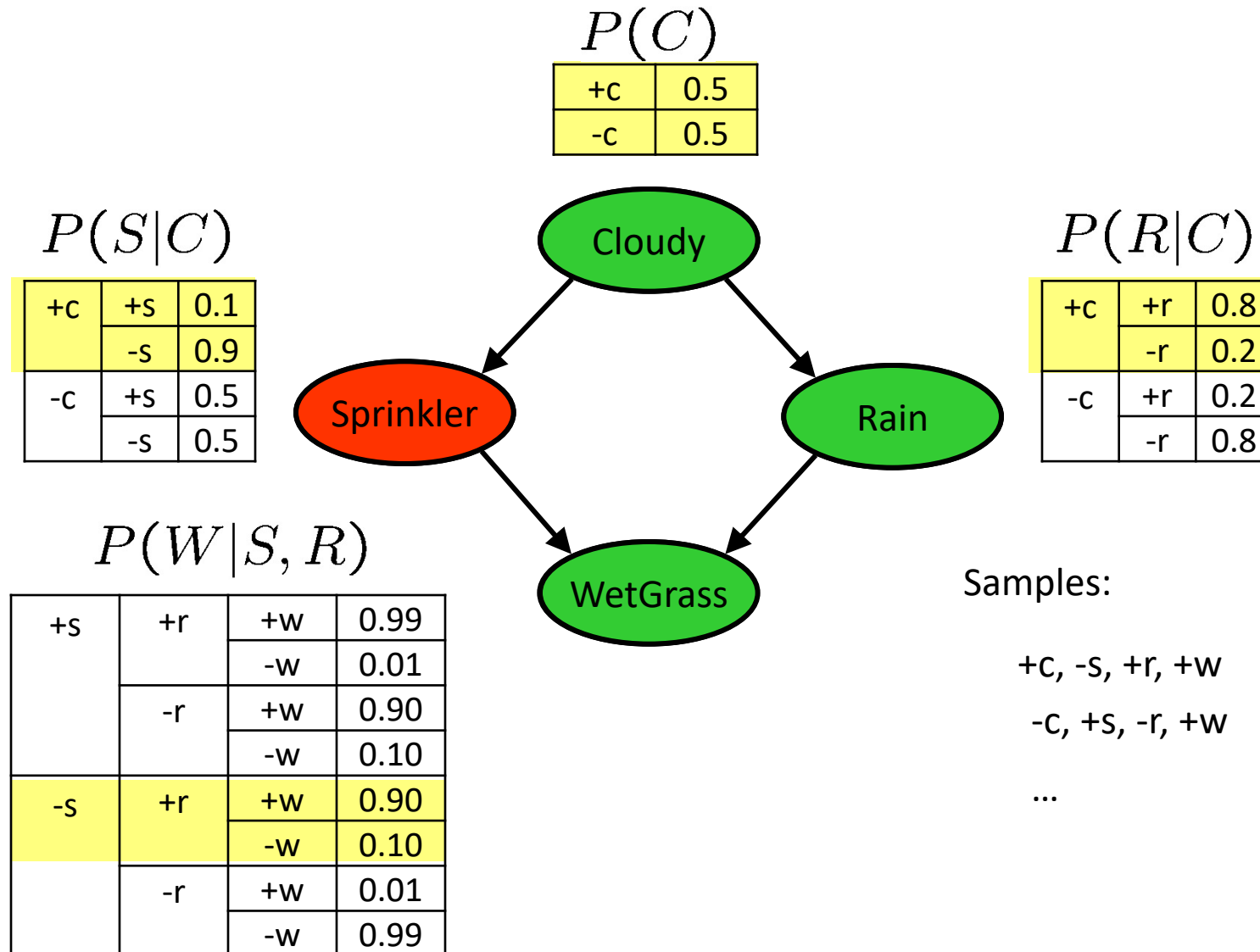
Sampling in Bayes' Nets

- Prior Sampling
- Rejection Sampling
- Likelihood Weighting
- Gibbs Sampling

Prior Sampling

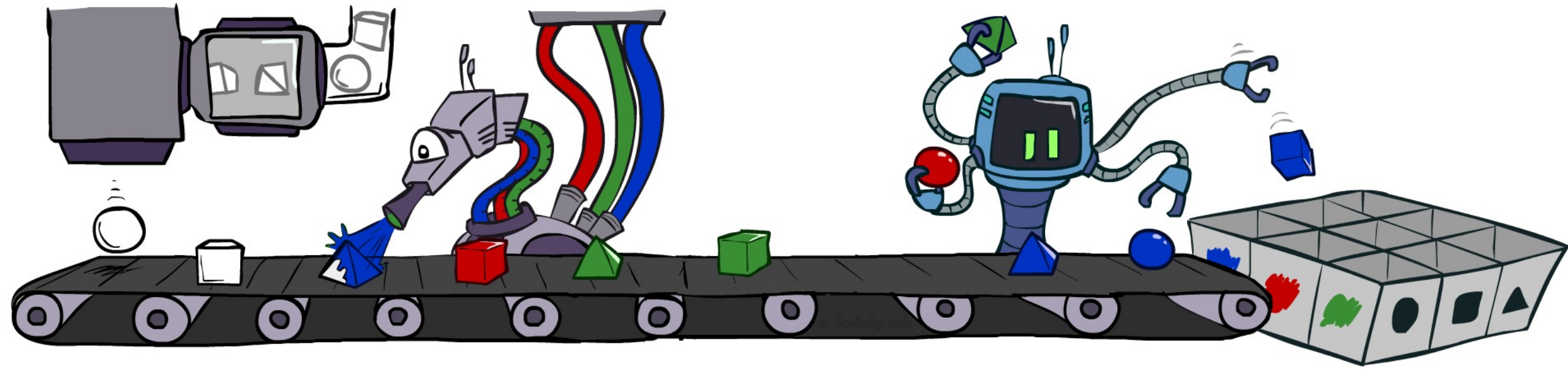


Prior Sampling



Prior Sampling

- For $i=1, 2, \dots, n$
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- Return $(x_1, x_2, \dots, x_n)$



Prior Sampling

- This process generates samples with probability:

$$S_{PS}(x_1 \dots x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(X_i)) = P(x_1 \dots x_n)$$

...i.e. the BN's joint probability

- Let the number of samples of an event be $N_{PS}(x_1 \dots x_n)$
- Then
$$\begin{aligned} \lim_{N \rightarrow \infty} \hat{P}(x_1, \dots, x_n) &= \lim_{N \rightarrow \infty} N_{PS}(x_1, \dots, x_n) / N \\ &= S_{PS}(x_1, \dots, x_n) \\ &= P(x_1 \dots x_n) \end{aligned}$$
- I.e., the sampling procedure is **consistent**

Example

- We'll get a bunch of samples from the BN:

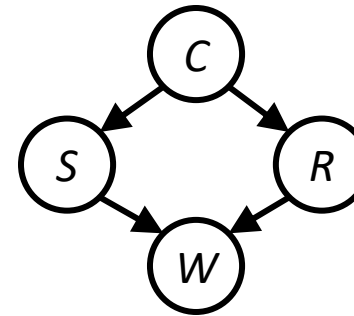
+c, -s, +r, +w

+c, +s, +r, +w

-c, +s, +r, -w

+c, -s, +r, +w

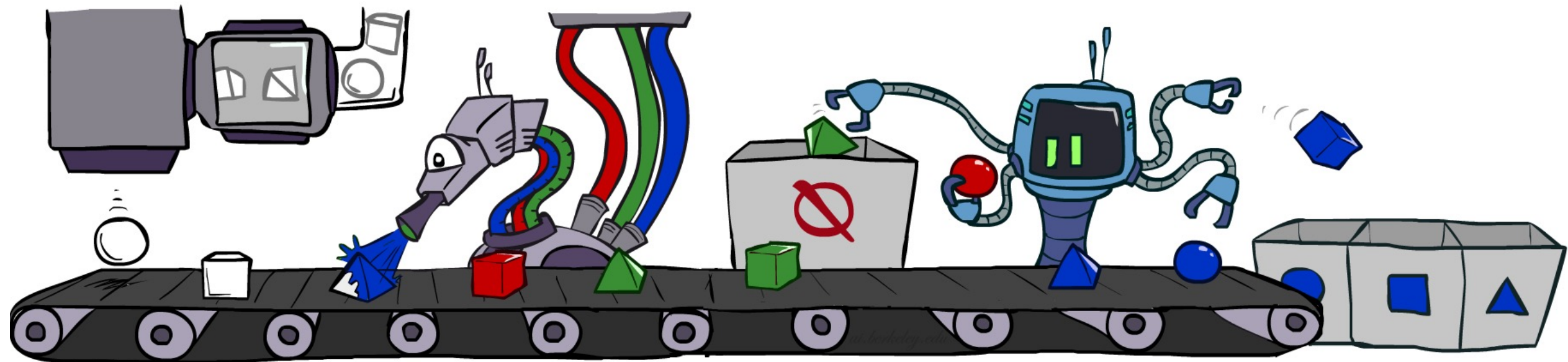
-c, -s, -r, +w



- If we want to know $P(W)$

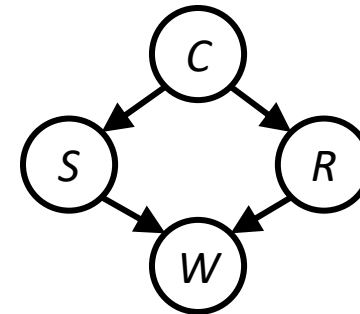
- We have counts $\langle +w:4, -w:1 \rangle$
- Normalize to get $P(W) = \langle +w:0.8, -w:0.2 \rangle$
- This will get closer to the true distribution with more samples
- Can estimate anything else, too
- What about $P(C \mid +w)$? $P(C \mid +r, +w)$? $P(C \mid -r, -w)$?
- Fast: can use fewer samples if less time (what's the drawback?)

Rejection Sampling



Rejection Sampling

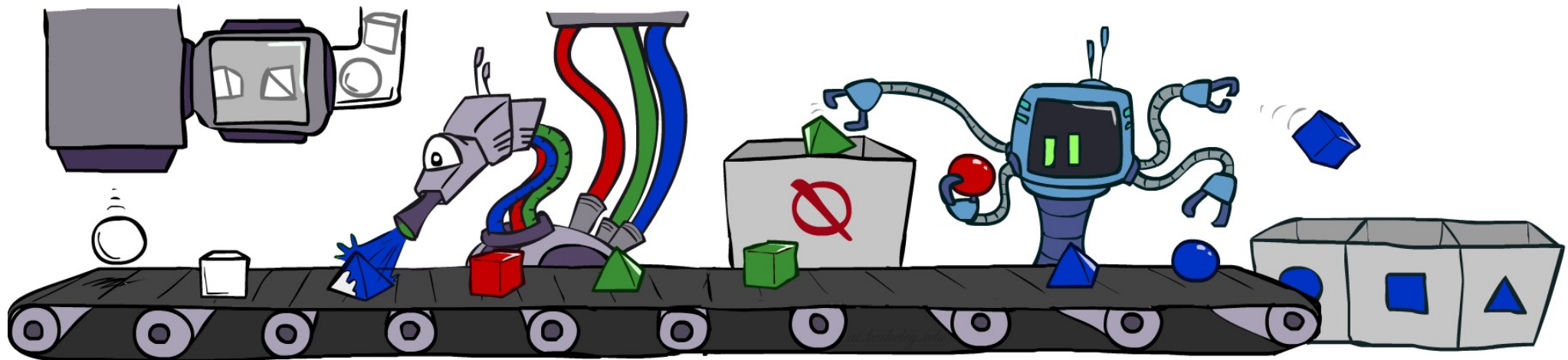
- Let's say we want $P(C)$
 - No point keeping all samples around
 - Just tally counts of C as we go
- Let's say we want $P(C \mid +s)$
 - Same thing: tally C outcomes, but ignore (reject) samples which don't have $S=+s$
 - This is called rejection sampling
 - It is also consistent for conditional probabilities (i.e., correct in the limit)



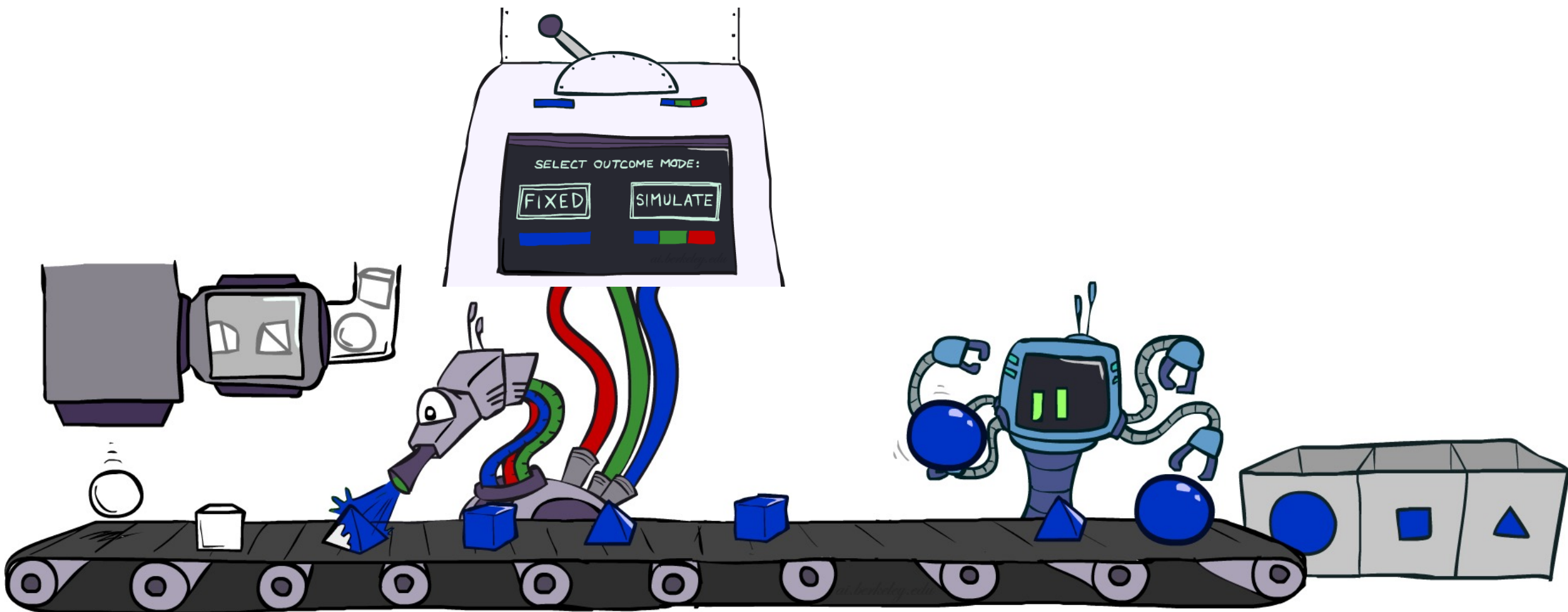
+c, -s, +r, +w
+c, +s, +r, +w
-c, +s, +r, -w
+c, -s, +r, +w
-c, -s, -r, +w

Rejection Sampling

- IN: evidence instantiation
- For $i=1, 2, \dots, n$
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
 - If x_i not consistent with evidence
 - Reject: Return, and no sample is generated in this cycle
- Return $(x_1, x_2, \dots, x_n)$

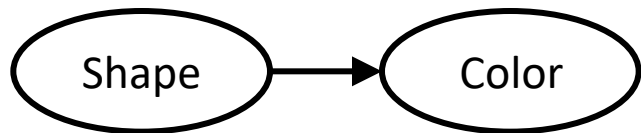


Likelihood Weighting

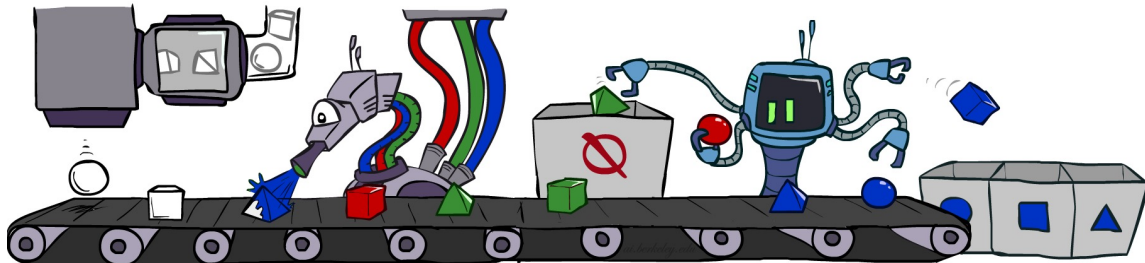


Likelihood Weighting

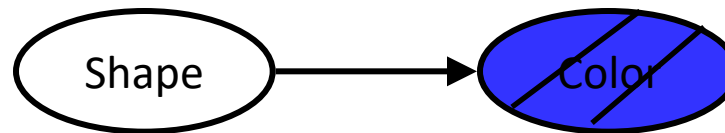
- Problem with rejection sampling:
 - If evidence is unlikely, rejects lots of samples
 - Evidence not exploited as you sample
 - Consider $P(\text{Shape} | \text{blue})$



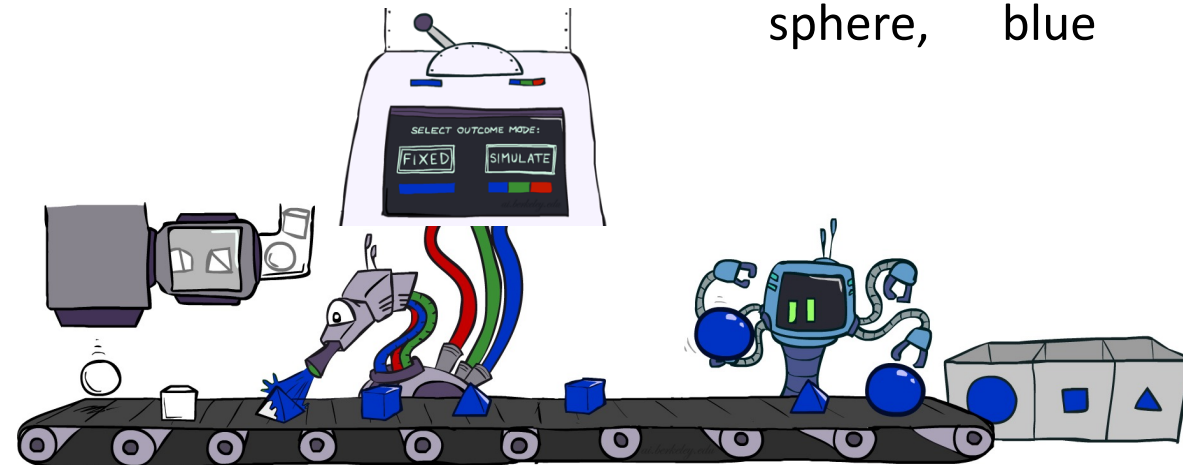
pyramid, green
pyramid, red
sphere, blue
cube, red
~~sphere, green~~



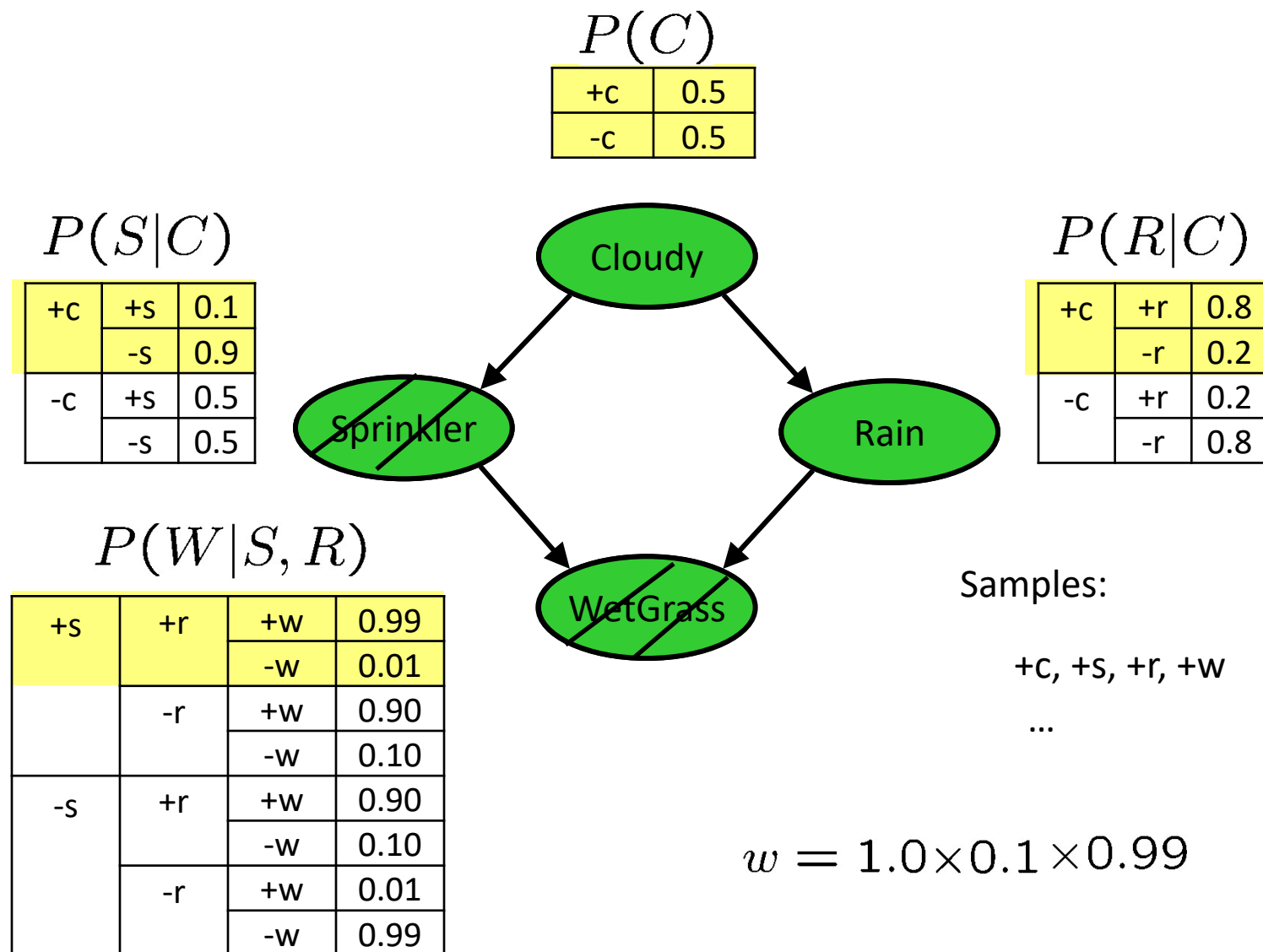
- Idea: fix evidence variables and sample the rest
 - Problem: sample distribution not consistent!
 - Solution: weight by probability of evidence given parents



pyramid, blue
pyramid, blue
sphere, blue
cube, blue
sphere, blue

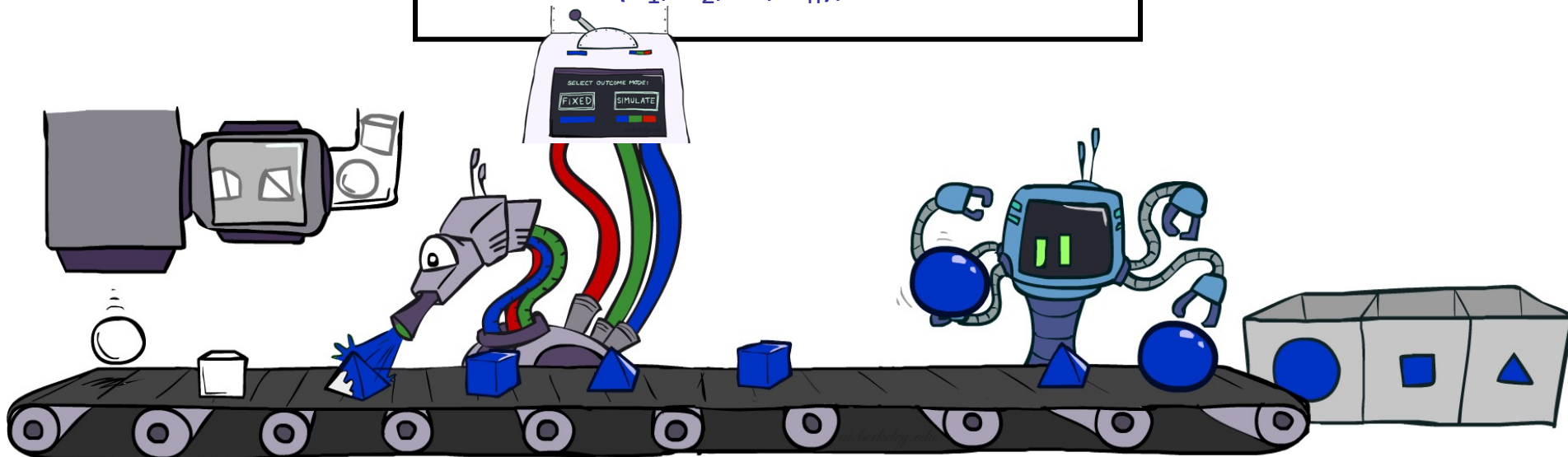


Likelihood Weighting



Likelihood Weighting

- IN: evidence instantiation
- $w = 1.0$
- for $i=1, 2, \dots, n$
 - if X_i is an evidence variable
 - $X_i = \text{observation } x_i \text{ for } X_i$
 - Set $w = w * P(x_i \mid \text{Parents}(X_i))$
 - else
 - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- return $(x_1, x_2, \dots, x_n), w$



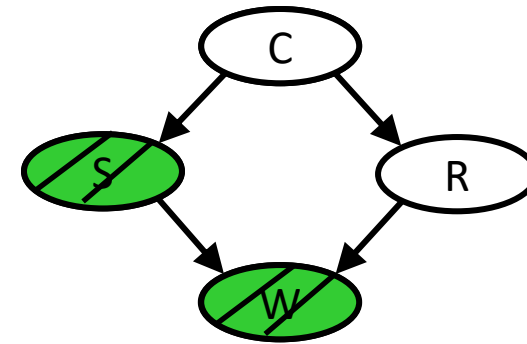
Likelihood Weighting

- Sampling distribution if z sampled and e fixed evidence

$$S_{WS}(z, e) = \prod_{i=1}^l P(z_i | \text{Parents}(Z_i))$$

- Now, samples have weights

$$w(z, e) = \prod_{i=1}^m P(e_i | \text{Parents}(E_i))$$

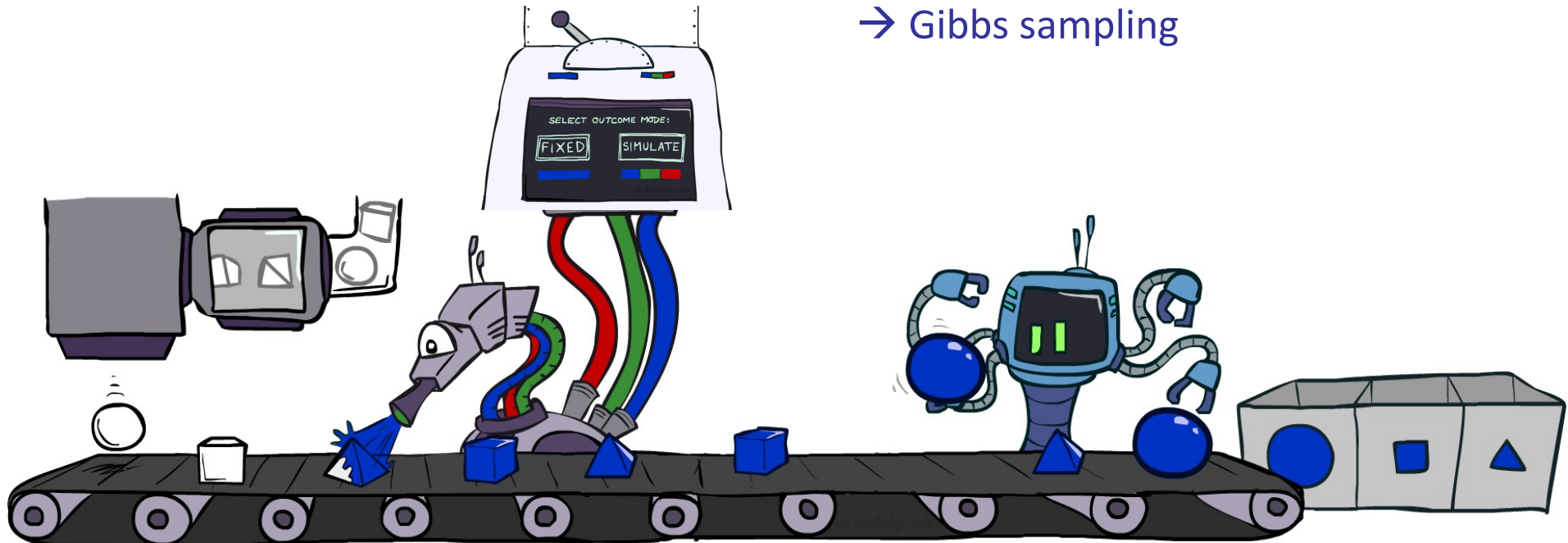


- Together, weighted sampling distribution is consistent

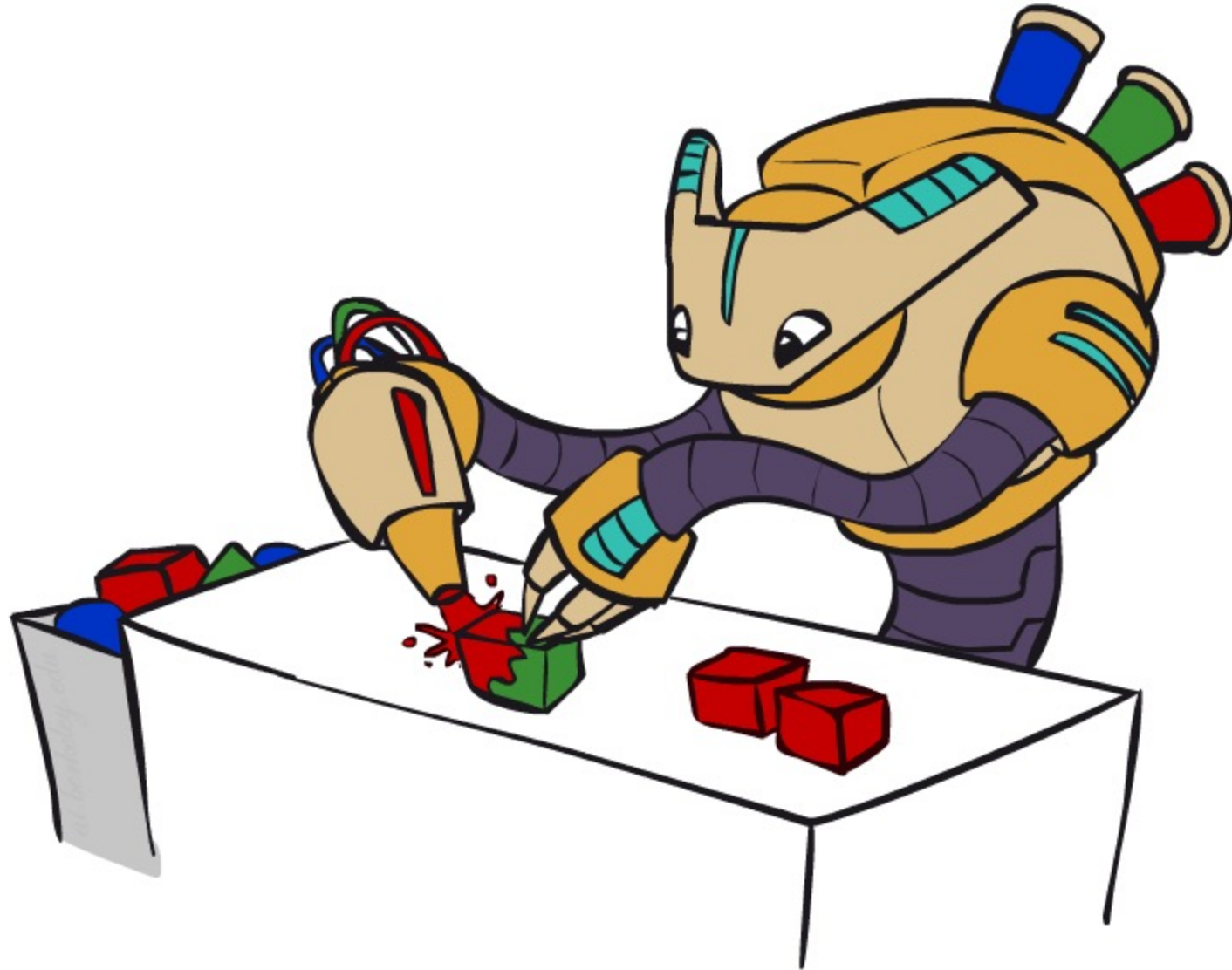
$$\begin{aligned} S_{WS}(z, e) \cdot w(z, e) &= \prod_{i=1}^l P(z_i | \text{Parents}(z_i)) \prod_{i=1}^m P(e_i | \text{Parents}(e_i)) \\ &= P(z, e) \end{aligned}$$

Likelihood Weighting

- Likelihood weighting is good
 - We have taken evidence into account as we generate the sample
 - E.g. here, W 's value will get picked based on the evidence values of S , R
 - More of our samples will reflect the state of the world suggested by the evidence
- Likelihood weighting doesn't solve all our problems
 - Evidence influences the choice of downstream variables, but not upstream ones (C isn't more likely to get a value matching the evidence)
- We would like to consider evidence when we sample every variable
 - Gibbs sampling



Gibbs Sampling



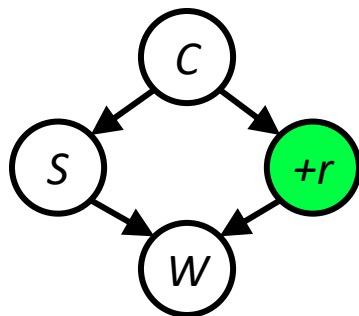
Gibbs Sampling

- *Procedure:* keep track of a full instantiation $x_1, x_2, \dots, x_n$. Start with an arbitrary instantiation consistent with the evidence. Sample one variable at a time, conditioned on all the rest, but keep evidence fixed. Keep repeating this for a long time.
- *Property:* in the limit of repeating this infinitely many times the resulting sample is coming from the correct distribution
- *Rationale:* both upstream and downstream variables condition on evidence.
- In contrast: likelihood weighting only conditions on upstream evidence, and hence weights obtained in likelihood weighting can sometimes be very small. Sum of weights over all samples is indicative of how many “effective” samples were obtained, so want high weight.

Gibbs Sampling Example: $P(S \mid +r)$

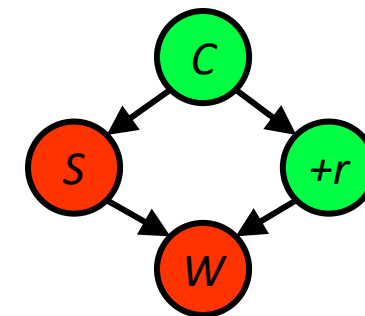
- Step 1: Fix evidence

- $R = +r$



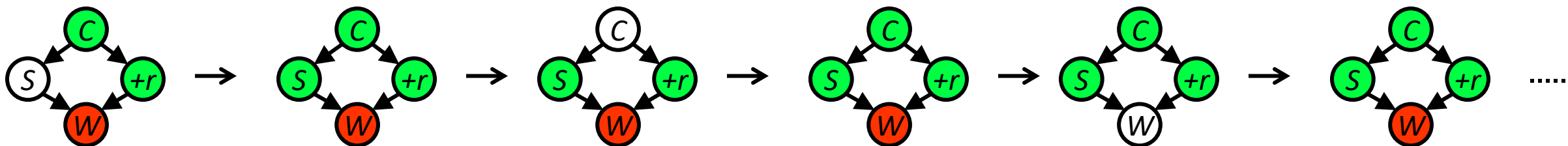
- Step 2: Initialize other variables

- Randomly



- Steps 3: Repeat

- Choose a non-evidence variable X
 - Resample X from $P(X \mid \text{all other variables})$



Sample from $P(S \mid +c, -w, +r)$

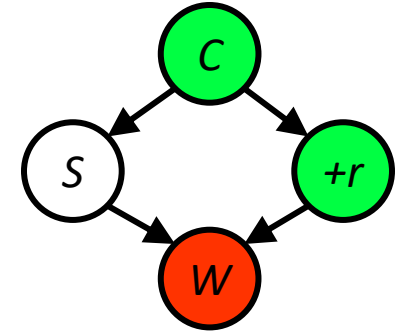
Sample from $P(C \mid +s, -w, +r)$

Sample from $P(W \mid +s, +c, +r)$

Efficient Resampling of One Variable

- Sample from $P(S \mid +c, +r, -w)$

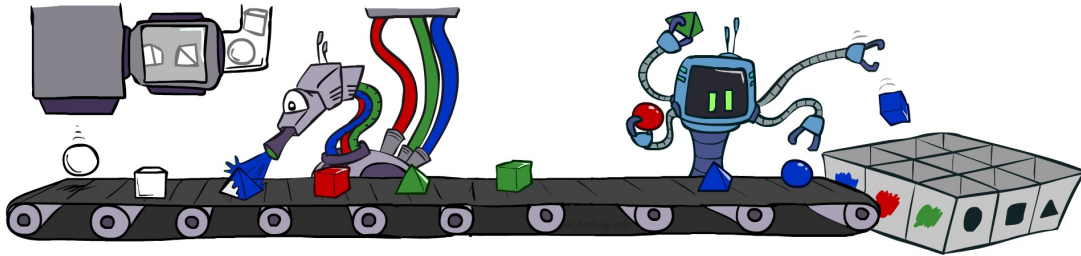
$$\begin{aligned} P(S \mid +c, +r, -w) &= \frac{P(S, +c, +r, -w)}{P(+c, +r, -w)} \\ &= \frac{P(S, +c, +r, -w)}{\sum_s P(s, +c, +r, -w)} \\ &= \frac{P(+c)P(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{\sum_s P(+c)P(s \mid +c)P(+r \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(+c)P(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{P(+c)P(+r \mid +c) \sum_s P(s \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(S \mid +c)P(-w \mid S, +r)}{\sum_s P(s \mid +c)P(-w \mid s, +r)} \end{aligned}$$



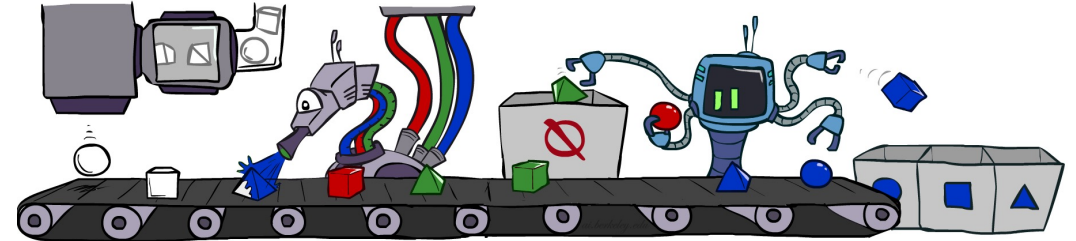
- Many things cancel out – only CPTs with S remain!
- More generally: only CPTs that have resampled variable need to be considered, and joined together

Bayes' Net Sampling Summary

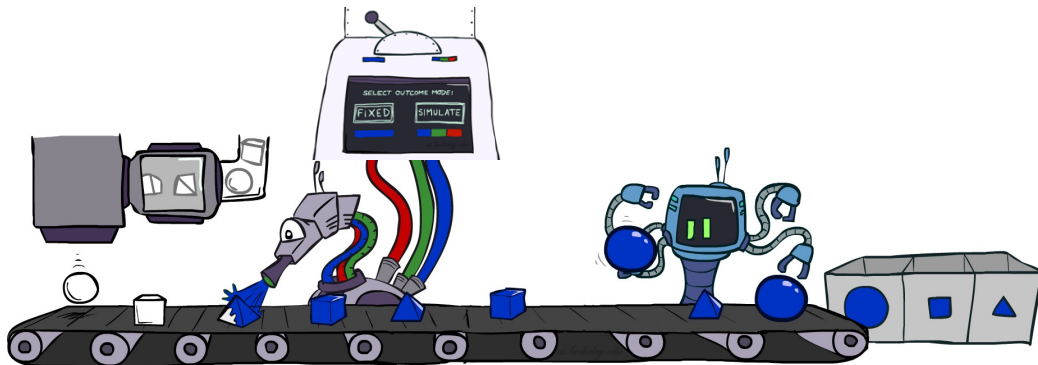
- Prior Sampling P



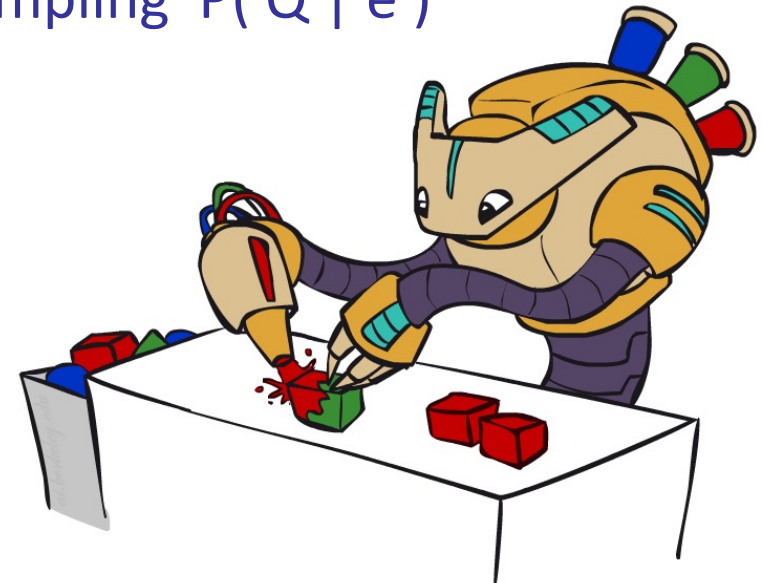
- Rejection Sampling $P(Q | e)$



- Likelihood Weighting $P(Q | e)$



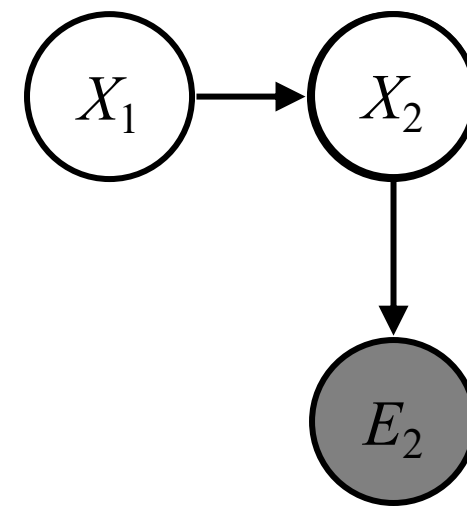
- Gibbs Sampling $P(Q | e)$



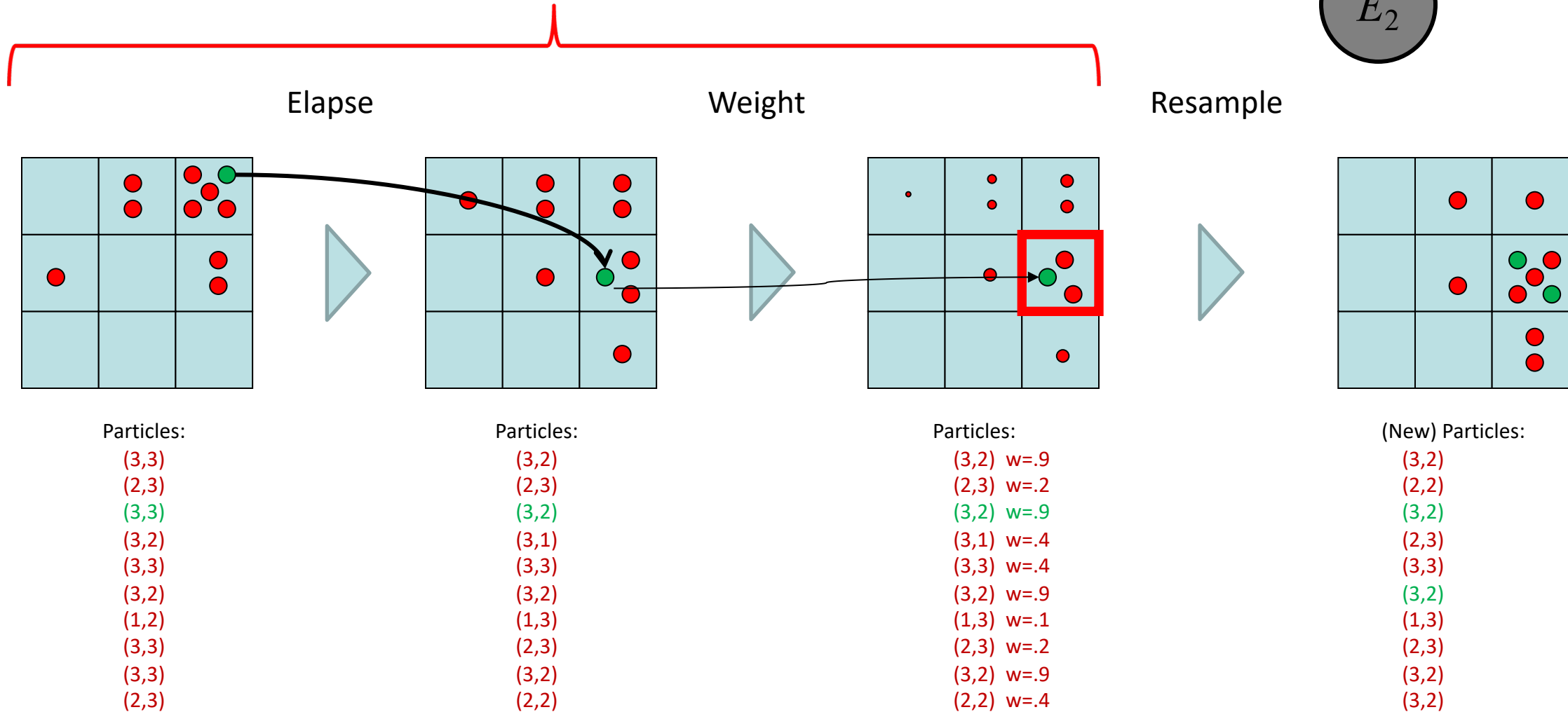
Further Reading on Gibbs Sampling*

- Gibbs sampling produces sample from the query distribution $P(Q | e)$ in limit of re-sampling infinitely often
- Gibbs sampling is a special case of more general methods called Markov chain Monte Carlo (MCMC) methods
 - Metropolis-Hastings is one of the more famous MCMC methods (in fact, Gibbs sampling is a special case of Metropolis-Hastings)
- You may read about Monte Carlo methods – they're just sampling

How About Particle Filtering?



= likelihood weighting



Particle Filtering

- Particle filtering operates on ensemble of samples
 - Performs likelihood weighting for each individual sample to elapse time and incorporate evidence
 - Resamples from the weighted ensemble of samples to focus computation for the next time step where most of the probability mass is estimated to be